

This session we're going to cover:

- Boolean operations
- If/elif/else condition

```
In [1]: # بیایید نگاهی به چند عملگر بولی (Boolean operations) بیندازیم
a = 13
b = 33
print(a > b)
print(a < b)
print(a == b)
print(a != b)
print(a >= b)
print(a <= b)
```

False
True
False
True
False
True

Python Comparison Operators

Operator	Meaning	Example
<code>==</code>	Is Equal To	<code>3 == 5</code> gives us <code>False</code>
<code>!=</code>	Not Equal To	<code>3 != 5</code> gives us <code>True</code>
<code>></code>	Greater Than	<code>3 > 5</code> gives us <code>False</code>
<code><</code>	Less Than	<code>3 < 5</code> gives us <code>True</code>
<code>>=</code>	Greater Than or Equal To	<code>3 >= 5</code> give us <code>False</code>
<code><=</code>	Less Than or Equal To	<code>3 <= 5</code> gives us <code>True</code>

دستور if...else در پایتون

در پایتون دستور `if`

به ما امکان می‌دهد تا بر اساس یک شرط مشخص، بلوکی از کد را اجرا کنیم یا از آن صرف‌نظر نماییم

این قابلیت به ما اجازه می‌دهد برنامه‌هایی بسازیم که بسته به شرایط، واکنش‌های متفاوتی نشان می‌دهند

```
In [1]: age = 20
if age >= 18:
    print("Eligible to vote.")
```

Eligible to vote.

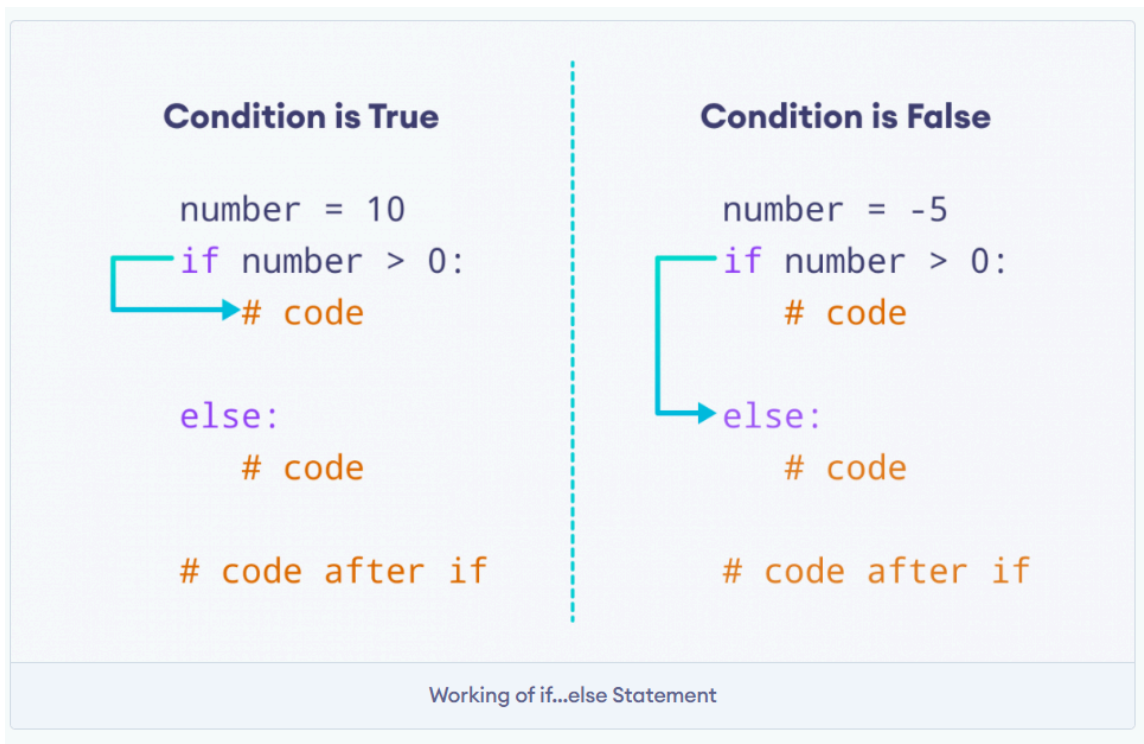
تورفتگی در پایتون (Indentation in Python)

پایتون برای تعریف یک بلوک کد، مانند بدنه دستور `if` از تورفتگی استفاده می‌کند

Indentation is 4 space or 1 tab

```
In [3]: age = 10
if age <= 12:
    print("Travel for free.")
else:
    print("Pay for ticket.")
```

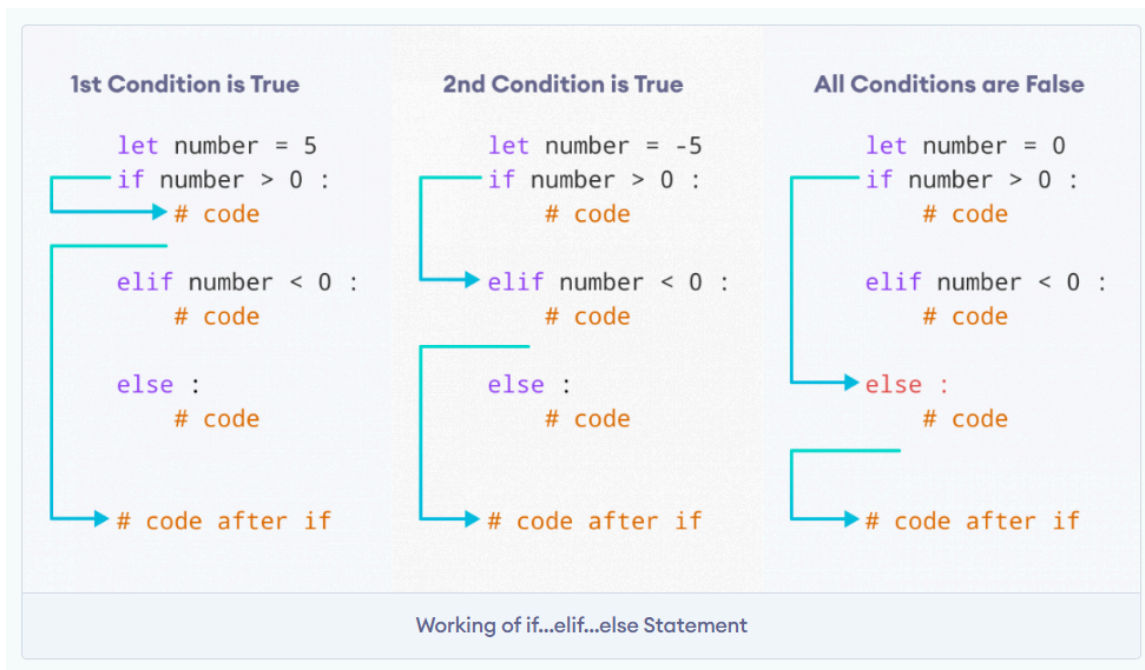
Travel for free.



<https://www.programiz.com/python-programming/if-elif-else>

```
In [4]: age = 25
if age <= 12:
    print("Child.")
elif age <= 19:
    print("Teenager.")
elif age <= 35:
    print("Young adult.")
else:
    print("Adult.")
```

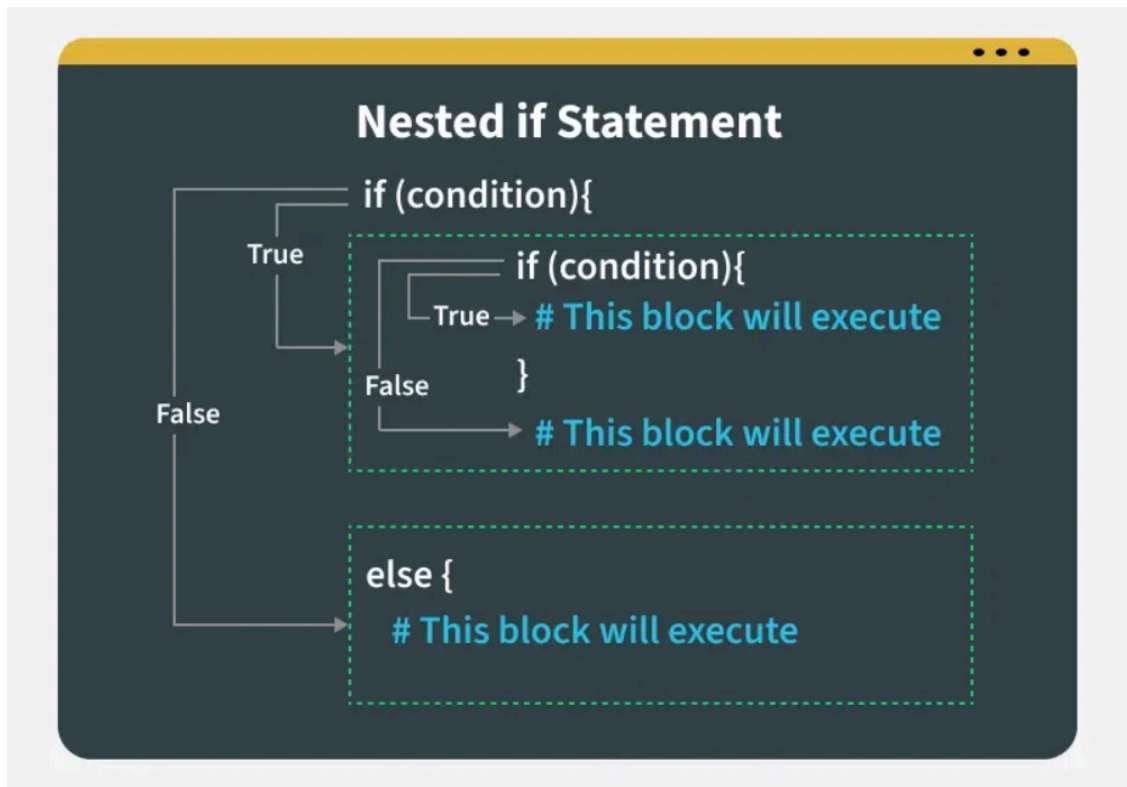
Young adult.



<https://www.programiz.com/python-programming/if-elif-else>

```
In [5]: age = 70
is_member = True
if age >= 60:
    if is_member:
        print("30% senior discount!")
    else:
        print("20% senior discount.")
else:
    print("Not eligible for a senior discount.")
```

30% senior discount!



<https://www.geeksforgeeks.org/python/python-if-else/>

Ternary Conditional Statement

```
In [6]: age = 20
s = "Adult" if age >= 18 else "Minor"
print(s)
```

Adult

مثال: منطق احراز هویت کاربر با استفاده از `if...else`

```
In [3]: # Username and password stored in database
username_db = "admin"
password_db = "sparrow@123"

# Username and password entered by the user
username = input("Enter username: ")
password = input("Enter password: ")

# Check if username & password in database matches user's input
if (username == username_db) and (password == password_db):
    print("Welcome back.")
else:
    print("Access denied.")
```

Welcome back.

Python Logical Operators

Operator	Example	Meaning
<code>and</code>	<code>a and b</code>	Logical AND: <code>True</code> only if both the operands are <code>True</code>
<code>or</code>	<code>a or b</code>	Logical OR: <code>True</code> if at least one of the operands is <code>True</code>
<code>not</code>	<code>not a</code>	Logical NOT: <code>True</code> if the operand is <code>False</code> and vice-versa.

mini project

Quiz game

```
In [1]: print('welcome to quiz game')

playing = input('Do you want to play? ')

if playing != 'yes':
    quit()

print("Ok let's play :) ")
score = 0

answer = input('What does CPU stand for? ')
if answer == 'central processing unit':
    print('Correct')
    score += 1
else:
    print('Incorrect!')

answer = input('What does GPU stand for? ')
if answer == 'graphics processing unit':
    print('Correct')
    score += 1
else:
    print('Incorrect!')

answer = input('What does RAM stand for? ')
if answer == 'random access memory':
    print('Correct')
    score += 1
else:
    print('Incorrect!')

answer = input('What does PSU stand for? ')
if answer == 'power supply':
```

```
    print('Correct')
    score += 1
else:
    print('Incorrect!')

print(f'you got {score} questions correct.')
print(f'you got {(score / 4 * 100)}%')
```